

State-Action Information Leakage in Sepsis Reinforcement Learning: An Empirical Audit of SOFA-Derived Clinical State Representations in MIMIC-IV

DRAFT FOR INTERNAL REVIEW — NOT FOR SUBMISSION OR DISTRIBUTION

Short title: SOFA Leakage Audit in Sepsis RL

Authors:

Abstract

Background/Objectives

Offline reinforcement learning (RL) has been proposed as a means of learning individualized sepsis treatment policies from retrospective intensive care unit (ICU) data, most notably using the Sequential Organ Failure Assessment (SOFA) score, or its components, as part of the patient state representation (Komorowski et al., 2018). Because the SOFA cardiovascular subscore is partly a function of vasopressor administration, and because clinical features are often aggregated over windows that may overlap periods of prior treatment, a state representation built in this way may encode information about the treatment action itself rather than reflecting only pre-decision patient physiology. This study conducts an empirical methodological audit, using MIMIC-IV, to characterize two distinct mechanisms by which treatment information may enter a nominally pre-decision clinical state: (1) definitional/construction-based leakage arising from the mathematical definition of the SOFA cardiovascular subscore, and (2) temporal/window-overlap leakage arising from aggregation windows that span periods of prior treatment.

Methods

We extracted an adult, first-ICU-stay, Sepsis-3-qualifying cohort with vasopressor exposure from MIMIC-IV (11,354 ICU stays; 7,497,914 vitals/labs/FiO₂ rows; 5,428,937 vasopressor dose events). SOFA cardiovascular subscores were decomposed into mean arterial pressure (MAP)-attributable and vasopressor dose-attributable components under two aggregation windows: a 4-hour decision-aligned window (280,837 scored timesteps) and a 24-hour conventional window (326,647 scored timesteps). We quantified (Experiment 1) the proportion of timesteps whose cardiovascular subscore was dose-determined; (Experiment 2) the recoverability of the next-bin vasopressor action (positive rate 44.6%; $n = 170,299$ modeling timesteps) from five nested state variants (A: full conventional state; B: without total SOFA; C: without cardiovascular SOFA; D: treatment-decomposed state; E: physiology-only state) using logistic regression, random forest, gradient boosting, and mutual information, with patient-grouped cross-validation; and (Experiment 3) the fraction of 24-hour-window decision points whose feature-aggregation window overlapped a prior treatment period.

Results

Under the 4-hour window, 30.5% of all eligible timesteps (100% of vasopressor-treated timesteps) were dose-determined, with a mean dose-attributable cardiovascular SOFA contribution of 1.05 points; under the 24-hour window, the corresponding figures were 26.3% and a mean contribution of 0.90 points. Action-recoverability probes showed area under the receiver operating characteristic curve (AUROC) ordering $A \approx B (0.869-0.900) > C (0.845-0.885) > D \approx E (0.727-0.791)$ across all three probe families, with mutual information following the same ordering ($0.031 \rightarrow 0.017$). Among 204,372 24-hour-window decision points, the mean contaminated-window fraction was 56.1% (median 100%; 25th percentile 0%), and 55.7% of decision points had more than 50% overlap with a prior treatment period. The corresponding 4-hour-window overlap analysis and the pre-specified negative-control battery (label shuffling, temporal-offset probes, prevalence-matched placebo features) were not completed at the time of this report.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert clustered-bootstrap 95% confidence intervals for all reported AUROC and mutual-information values.]

Conclusions

A substantial minority to majority of SOFA-scored timesteps in this cohort had cardiovascular subscores mechanically determined by vasopressor dosing rather than MAP alone, and next-bin vasopressor action was substantially more recoverable from conventional state representations than from treatment-decomposed or physiology-only representations. These findings are consistent with, but do not by themselves establish the magnitude of, treatment-information embedding in SOFA-based clinical states used for offline sepsis RL; the observed AUROC gap should be treated as an upper-bound estimate pending completion of negative-control analyses that can help separate leakage from treatment persistence, autocorrelation, and legitimate clinical predictability. Investigators constructing SOFA-based states or rewards for sepsis RL should consider decomposed or lagged alternatives and should report window-overlap diagnostics as a matter of course.

Keywords

Offline reinforcement learning; sepsis; SOFA score; state representation; information leakage; target leakage; MIMIC-IV; clinical decision support; causal inference; temporal aggregation

1. Introduction

Sepsis remains a leading cause of morbidity and mortality in intensive care units (ICUs) worldwide, and its management particularly the titration of intravenous fluids and vasopressors involves sequential decisions made under uncertainty, with substantial variation in practice across clinicians and institutions (Singer et al., 2016). This combination of high stakes, sequential structure, and practice variation has made sepsis a focal application area for offline reinforcement learning (RL), a family of methods that aims to learn treatment policies from retrospective observational data without requiring new clinical trials. The most widely cited example, the “AI Clinician,” used a discretized state space derived substantially from SOFA components together with a discretized vasopressor/fluid action space, trained on retrospective ICU data (Komorowski et al., 2018). Since that work, a growing literature has proposed related architectures for sepsis treatment-policy learning, generally sharing the assumption that a sufficiently rich set of vital signs, laboratory values, and severity scores can serve as a valid Markov state for the RL problem.

The validity of offline RL for high-stakes clinical decisions depends critically on the validity of the state representation. In the standard Markov decision process (MDP) formulation underlying most offline RL methods, the state at a given decision point is assumed to summarize the patient's condition *prior* to the action being modeled, so that the learned policy reflects a mapping from pre-decision information to a treatment recommendation. If the state representation instead contains information that is a direct or indirect function of the action itself — whether the current action or a temporally proximate prior instance of it — the resulting policy-learning and policy-evaluation procedures can be compromised in ways that are difficult to detect from performance metrics alone. This concern is closely related to, but conceptually distinct from, the general phenomenon of *leakage* in predictive modeling, defined by Kaufman et al. (2012) as the introduction of information about a modeling target that would not legitimately be available at the time a prediction is required to be made.

The Sequential Organ Failure Assessment (SOFA) score is a natural point of focus for this concern. Originally developed to describe the degree of organ dysfunction in critically ill patients (Vincent et al., 1996), and later incorporated into the Sepsis-3 consensus definitions as a basis for identifying sepsis-related organ dysfunction (Singer et al., 2016), SOFA has become one of the most widely used severity measures in both sepsis research and, subsequently, sepsis RL. Its cardiovascular subscore, however, is not a pure physiological severity measure: by construction, it is scored using a combination of mean arterial pressure (MAP) and the type/dose of vasopressor or inotropic support a patient is receiving, such that a patient already receiving vasopressors is assigned an elevated cardiovascular subscore in part *because* of that treatment, independent of their current MAP. When vasopressor administration is simultaneously the action being modeled in an RL formulation, a SOFA-derived state that includes the cardiovascular subscore may therefore encode information about the treatment action through the scoring definition itself — a mechanism we term **definitional/construction-based leakage**.

A second, conceptually distinct mechanism arises from how clinical features are aggregated over time. Many implementations compute a patient's state features — including SOFA — using a retrospective window (e.g., the most recent 24 hours) preceding a decision point. If that window overlaps a period during which a prior vasopressor treatment was already active, the resulting state may reflect prior

treatment status or treatment-mediated physiology (e.g., a MAP that has already been raised by vasopressors administered earlier in the window) even where no scoring rule directly encodes treatment. We term this **temporal/window-overlap leakage**. Unlike definitional leakage, this mechanism does not require that the state literally contain a treatment-derived score component; it can arise purely from the interaction between aggregation-window length, decision-interval length, and the autocorrelation of treatment over time.

These two mechanisms must, in turn, be distinguished from several related but non-equivalent phenomena that also produce associations between clinical state and treatment action: the physiological *consequences* of previous treatment (e.g., a genuinely improved MAP caused by an earlier vasopressor dose, which is a real causal effect rather than an artifact of scoring or windowing); the ordinary *autocorrelation* or *persistence* of treatment decisions over time (clinicians who started a vasopressor recently are likely to continue it, independent of any state-construction artifact); the ordinary clinical *predictability* of treatment from severity (sicker patients legitimately receive more vasopressors, and a model detecting this is not thereby detecting a construction flaw); *confounding* by unmeasured factors that jointly influence treatment and outcome; and, more generally, *post-treatment conditioning* and violations of temporal ordering as understood in the causal-inference literature. Distinguishing these mechanisms is essential because they carry different implications and different remedies: definitional leakage can potentially be addressed by decomposing or omitting treatment-linked score components; temporal-overlap leakage can potentially be addressed by shortening aggregation windows or using strictly pre-treatment feature extraction; but ordinary clinical predictability and treatment persistence are not, by themselves, defects to be corrected, and a state representation that fails to predict the action at all would likely be failing to capture genuine illness severity.

Despite the centrality of SOFA to sepsis RL, we are not aware of a systematic, quantitative audit — conducted directly on patient-level ICU data — of how much of SOFA-based action recoverability is attributable to definitional construction, how much to temporal window overlap, and how much remains after both are addressed. Existing methodological guidance for RL in healthcare (Gottesman et al., 2019) emphasizes the general importance of valid state and action definitions but does not provide an empirical, SOFA-specific accounting of this kind. This gap matters because SOFA-derived states and SOFA-derived rewards are used not only in the AI Clinician but across a broader family of sepsis-RL studies, and because published performance metrics for such policies (e.g., estimated policy value, off-policy evaluation results) can be difficult to interpret without knowing whether the underlying state representation was already informative about the action being evaluated.

Study aims. This study has four aims: (1) to formally distinguish definitional/construction-based leakage from temporal/window-overlap leakage in SOFA-derived clinical states, using explicit, technically defensible terminology; (2) to quantify, in a MIMIC-IV sepsis cohort, the proportion of SOFA-scored timesteps whose cardiovascular subscore is mechanically determined by vasopressor dosing rather than MAP, under both a 4-hour decision-aligned and a 24-hour conventional aggregation window; (3) to quantify how recoverable the next-bin vasopressor action is from nested state representations ranging from a full conventional state to a physiology-only, treatment-decomposed state, using multiple probe-model families; and (4) to quantify, for the 24-hour window, the extent to which feature-aggregation

windows overlap periods of prior treatment. Consistent with the exploratory and partially completed nature of this audit, we report negative-control analyses and a 4-hour temporal-overlap comparison as pre-specified but not yet completed, and we explicitly avoid characterizing the observed action-recoverability gap as a definitive, quantified causal effect of leakage.

2. Related Work / Background

2.1 Offline Reinforcement Learning in Healthcare

Offline (or “batch”) RL methods learn a decision policy from a fixed dataset of previously collected trajectories, without further interaction with the environment during training. In healthcare, this property is attractive because it avoids the need for online experimentation with real patients, but it also means that the learned policy is only as trustworthy as the assumptions underlying the state, action, and reward definitions used to construct the training trajectories from observational electronic health record (EHR) data (Gottesman et al., 2019). General guidance for RL in healthcare has emphasized the importance of careful problem formulation, including attention to the Markov assumption, the plausibility of the action space, and the interpretability of any evaluation metric used to compare a learned policy to observed clinician behavior (Gottesman et al., 2019).

2.2 Sepsis Treatment-Policy Learning

The most widely cited application of offline RL to sepsis treatment is the “AI Clinician,” which used retrospective MIMIC-III data to learn a policy over discretized intravenous fluid and vasopressor doses, using a state space substantially derived from vital signs, laboratory values, and severity indices including SOFA components, and reported that the value of the learned policy's recommendations exceeded that of observed clinician behavior on average (Komorowski et al., 2018). This work catalyzed a broader body of subsequent studies applying related offline RL architectures to sepsis treatment using MIMIC-III- or MIMIC-IV-derived cohorts.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert additional verified citations to subsequent sepsis-RL studies once the literature-audit table (Appendix A) has been populated from sources independently verified against primary publications; do not infer additional studies' methodological details without verification.]

Because the present audit's supplied source materials centered on the conceptual and empirical treatment of the AI Clinician-style formulation specifically, we note that the available methodological description in Komorowski et al. (2018) documents that a substantial portion of the state space was derived from SOFA and related severity indices, and that vasopressor dosing was part of the action space; the published methods description does not, by itself, permit exclusion of either definitional or temporal-overlap leakage as understood in this manuscript, nor does it establish that either mechanism materially affected that study's specific reported results. A dedicated, study-by-study assessment is presented in the literature-audit table (Appendix A) using the cautious certainty categories specified in our methodology.

2.3 State Representation in Clinical Reinforcement Learning

The choice of state representation determines what information is, and is not, available to an RL agent at each decision point. In principle, a valid state for a sequential decision problem should summarize all information relevant to the *future* consequences of a decision, without incorporating information that is only available *because* the decision (or a proximate instance of it) has already been made. In supervised learning, closely related concerns are addressed under the general heading of target leakage: information about the target variable that would not legitimately be available at prediction time and that inflates apparent model performance without corresponding real-world validity (Kaufman et al., 2012). The RL setting compounds this concern because the “target” being partially recoverable — the action — is not a fixed outcome label but is itself a component of the state-transition structure being modeled, and because RL policies are typically evaluated using off-policy methods that assume the state was constructed without reference to the action.

2.4 SOFA Construction and its Treatment-Linked Components

The SOFA score assigns points across six organ systems (respiratory, coagulation, hepatic, cardiovascular, renal, and neurological), with the cardiovascular subscore specifically defined using a combination of MAP thresholds and, at higher severity levels, the presence and dose of vasopressor or inotropic agents (Vincent et al., 1996). This construction means that, for a patient already receiving vasopressors, the cardiovascular subscore reflects treatment status in addition to (or, at some dose levels, instead of) directly measured hemodynamic severity. This is a well-established property of the score's design rather than a data-quality artifact, and it is not, by itself, a criticism of SOFA's validity for its original purpose of describing organ dysfunction over time; the concern raised in this manuscript is specific to the *repurposing* of SOFA or its subscores as an RL state feature under a treatment action that overlaps the scoring definition.

2.5 Target Leakage and Information Leakage

Leakage in predictive modeling has been formally treated as arising from a variety of mechanisms, including the inclusion of features that are proxies for, or direct functions of, the outcome being predicted, and features whose availability at “prediction time” differs from their availability in the deployed setting (Kaufman et al., 2012). Detection strategies proposed in this literature include examining feature-target correlations that are implausibly strong given domain knowledge, and testing model performance after removing suspect features — an approach closely related to the nested state-variant design (Variants A–E) used in Experiment 2 of the present study.

2.6 Post-Treatment Variables and Causal Inference

The causal-inference literature separately identifies the hazards of conditioning on variables measured after treatment has occurred, including so-called collider-stratification and post-treatment bias, which can distort estimated treatment effects even when no scoring-definition leakage is present.

[CITATION NEEDED — verify source supporting: a specific, primary causal-inference reference addressing post-treatment conditioning/collider bias, to be selected and verified before submission — e.g., a text or paper by Hernán and Robins, or Pearl, depending on which the author team intends to cite.]

Definitional and temporal leakage as characterized in this manuscript are best understood as specific, RL-relevant instances of the broader family of temporal-ordering violations this literature addresses, rather than as a novel causal phenomenon.

2.7 Temporal Aggregation in Longitudinal EHR Data

Electronic health record data are frequently aggregated into fixed time windows (e.g., “last 24 hours”) to construct tabular features for downstream modeling. This convention, while practically convenient, can obscure the exact timing relationship between a given feature value and any treatment administered nearby in time, particularly when the aggregation window is long relative to the decision interval.

[CITATION NEEDED — verify source supporting: primary methodological references specifically addressing temporal aggregation bias/window-overlap bias in longitudinal EHR modeling, to be selected and verified before submission.]

Experiment 3 of this study operationalizes this concern directly by quantifying window-overlap fractions rather than relying on general literature claims alone.

2.8 Methodological Guidance for RL in Healthcare

Existing guidance for applying RL to healthcare decision-making emphasizes several considerations directly relevant to this audit, including the importance of clearly specified state and action definitions and caution in interpreting policy-value estimates derived from observational data (Gottesman et al., 2019). This guidance is, to our knowledge, general in nature and does not provide SOFA-specific quantitative benchmarks of the kind this manuscript reports; the present study is intended to complement such guidance with a concrete, reproducible empirical audit.

3. Methods

3.1 Study Design

This study is an empirical methodological audit, not a treatment-effect or policy-evaluation study. Its purpose is to characterize, quantitatively, how much information about a treatment action (vasopressor administration) is embedded in a commonly used clinical state representation (SOFA-derived features) and through which of two candidate mechanisms — definitional construction or temporal window overlap — that information most plausibly enters. No treatment policy is proposed, trained, or evaluated in this study; Experiment 4 (policy-sensitivity analysis) was explicitly out of scope and was not conducted.

3.2 Data Source

Data were drawn from MIMIC-IV, a publicly available, de-identified electronic health record database sourced from Beth Israel Deaconess Medical Center and accessible to credentialed researchers via PhysioNet (Johnson et al., 2023).

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert exact MIMIC-IV version number (e.g., v2.2 or later), the PhysioNet DOI for that version, the date range of data extraction, the credentialed-access completion date/CITI training reference, and the SQL/ETL pipeline version or commit hash used for extraction.]

3.3 Cohort Definition

The analytic cohort was defined using the following inclusion criteria, applied in the order listed:

- Adult patients (age at admission meeting the adult threshold used by MIMIC-IV's de-identified age convention).
- First ICU stay only (subsequent ICU stays for the same hospitalization or patient were excluded).
- ICU length of stay (LOS) ≥ 1 day.
- Meeting Sepsis-3 criteria (suspected infection combined with an acute increase in SOFA score, per Singer et al., 2016).
- Evidence of vasopressor exposure during the ICU stay.

No exclusion criteria beyond these five are documented in the supplied source materials.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Confirm whether any additional exclusions were applied — e.g., exclusion of patients with extreme outlier vital-sign values, exclusion of stays with insufficient charting density, handling of comfort-care/withdrawal-of-care patients, or handling of patients missing key SOFA inputs (e.g., no creatinine measured). If none were applied, state so explicitly in the manuscript rather than leaving this ambiguous.]

This yielded 11,354 ICU stays as the base cohort.

3.4 Temporal Discretization and Decision Points

Patient trajectories were discretized into fixed decision intervals, with two aggregation-window conventions compared throughout the study: a 4-hour decision-aligned window and a 24-hour conventional window (the latter reflecting common practice in prior SOFA-based sepsis-RL literature, e.g., Komorowski et al., 2018). Each decision point corresponds to a bin boundary at which a next-bin vasopressor action label was assigned.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Confirm the exact binning implementation — whether bins are anchored to ICU admission time, to a fixed clock time, or to a rolling window; whether bins are non-overlapping or overlapping; how missing bins (no charted data in a given interval) were handled; and whether decision points were only generated during the vasopressor-eligible portion of the stay or across the entire ICU stay.]

3.5 State Construction

State features were constructed in three stages:

1. Native-resolution extraction. Vitals, laboratory values, and fraction of inspired oxygen (FiO₂) measurements were extracted at their original charted resolution (combined total: 7,497,914 rows across vitals/labs/FiO₂), together with vasopressor dose events (5,428,937 rows).

2. Aggregation. Native-resolution values were aggregated within each decision window (4-hour or 24-hour) using a summary statistic per feature.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Confirm the aggregation statistic(s) used per feature type — e.g., last-observed value, mean, worst value in window — since this choice affects both the substantive state construction and the interpretation of window-overlap results in Experiment 3.]

3. SOFA component computation. Aggregated values were used to compute the six SOFA subscores (respiratory, coagulation, hepatic, cardiovascular, renal, neurological) and total SOFA, separately under the 4-hour window (280,837 scored timesteps) and the 24-hour window (326,647 scored timesteps). Per the study's documented Limitations, the renal subscore was implemented using creatinine only (urine output was not incorporated).

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Confirm this is the only simplification relative to full SOFA scoring, and confirm the exact laboratory/vital-sign source tables and unit conventions used for each subscore.]

3.6 Action Definition

The action of interest was the *next-bin vasopressor action*: a label indicating whether vasopressor administration occurred in the decision interval immediately following a given state's window. Across the final Experiment 2 modeling table (170,299 timesteps), the positive rate for this action was 44.6%.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: The exact operationalization of this action is not documented in the supplied materials and must be specified precisely before submission, including: (a) whether the action is binary (any vasopressor vs. none) or multiclass/continuous (dose level, norepinephrine-equivalent conversion across agents); (b) the minimum duration or dose threshold, if any, required for a bin to be labeled positive; (c) how boundary cases are handled where a vasopressor infusion starts or stops partway through a bin; and (d) how multi-agent vasopressor regimens are aggregated into a single action label. This is flagged as essential rather than optional because the action definition materially affects the interpretation of the 44.6% positive rate and of all Experiment 2 recoverability results.]

3.7 SOFA Cardiovascular Decomposition

To isolate definitional leakage, the cardiovascular SOFA subscore at each timestep was decomposed into two components:

- MAP-attributable score: the cardiovascular subscore value that would be assigned using MAP thresholds alone, under the assumption of no vasopressor support.
- Dose-attributable score: the incremental cardiovascular subscore attributable specifically to vasopressor administration and dose, computed as the difference between the full (MAP + vasopressor) SOFA cardiovascular scoring rule's output and the MAP-attributable score.
- Final cardiovascular score: the standard SOFA cardiovascular subscore actually used in the state representation, equal to the maximum of the MAP-attributable and dose-attributable components under the conventional scoring rule (consistent with the original SOFA definition's use of vasopressor dose thresholds to assign higher point values than MAP alone would justify; Vincent et al., 1996).

A timestep was labeled **dose-determined** if the final cardiovascular score exceeded the MAP-attributable score — i.e., if vasopressor dosing, rather than measured MAP, was responsible for some or all of the assigned points.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert the exact MAP and dose thresholds used (standard SOFA cardiovascular scoring bands), the specific vasopressor agents and dose units included in the dose-attributable calculation, and confirmation of how missing MAP values within a window were handled for this decomposition.]

3.8 Experiment 1: Definitional Leakage Audit

For each aggregation window (4-hour, 24-hour) and each eligible timestep, we computed: (a) the proportion of all eligible timesteps labeled dose-determined; (b) the proportion of *vasopressor-treated* timesteps labeled dose-determined; and (c) the mean dose-attributable cardiovascular SOFA contribution (points), averaged across dose-determined timesteps. Consistent with the pre-specified interpretation guidance for this study, the population-level proportions in (a) are treated as the substantive findings of this experiment, while the near-ceiling proportion in (b) is treated primarily as a validation of the

decomposition's internal consistency (since, by construction, a treated patient's cardiovascular score is mechanically influenced by dose whenever the dose-based scoring band exceeds the MAP-based band).

3.9 Experiment 2: Action Recoverability

Experiment 2 assessed how recoverable the next-bin vasopressor action (Section 3.6) was from five nested state variants, constructed by progressively removing treatment-linked components from the 24-hour conventional state:

Variant	Description
A	Full conventional state (all vitals, labs, FiO2, and complete SOFA, including total and all subscores)
B	Full conventional state without total SOFA (subscores retained)
C	Full conventional state without total SOFA and without the cardiovascular SOFA subscore
D	Treatment-decomposed state: cardiovascular contribution restricted to the MAP-attributable component only (dose-attributable component removed), with other subscores retained
E	Physiology-only state: all directly treatment-linked scoring components removed, retaining only raw physiological measurements (vitals, labs, FiO2) and non-cardiovascular SOFA subscores computed without treatment-derived inputs

This nested design is a diagnostic tool for **information recoverability** — it measures how well an action label can be statistically predicted from a given feature set — and is explicitly not a clinical prediction model and not, by itself, proof of causal leakage. A high recoverability score indicates that the action is statistically predictable from the state; it does not by itself establish the *mechanism* (definitional construction, temporal overlap, treatment persistence, or legitimate severity-based predictability) responsible for that predictability. The comparison across nested variants is informative precisely because it isolates the marginal contribution of specific treatment-linked components (total SOFA between A and B; cardiovascular SOFA between B and C; the dose-attributable cardiovascular component between C and D) while holding all other features constant.

Three probe-model families were used: logistic regression, random forest, and gradient boosting, together with a non-model-based mutual information estimate between each state variant and the action label. All models were evaluated using patient-grouped cross-validation, so that no patient's timesteps appeared in both training and evaluation folds within a fold split.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert exact hyperparameters for each probe model (e.g., regularization strength for logistic regression; number of trees, max depth, and min-samples-leaf for random forest and gradient boosting); number of cross-validation folds and fold-assignment procedure; random seed(s) used; preprocessing steps (standardization, encoding); missing-data imputation method; class-weighting or resampling strategy given the 44.6% positive rate; and the number of bootstrap repetitions (if any) used to compute confidence intervals. Clustered-bootstrap 95% confidence intervals for all AUROC and mutual-information values are required before submission and are not yet available for insertion.]

3.10 Experiment 3: Temporal Window-Overlap Audit

For the 24-hour conventional window, we quantified, for each decision point, the **contaminated-window fraction**: the proportion of the 24-hour feature-aggregation window that temporally overlapped a period during which a prior vasopressor treatment was already active. Formally, for a decision point with aggregation window $[t - 24h, t)$ and a set of prior vasopressor-active intervals $\{[s_i, e_i]\}$, the contaminated-window fraction is:

A decision point was classified as having **majority overlap** if $f(t) > 0.5$. This audit was conducted over 204,372 decision points for the 24-hour window. The corresponding calculation for the 4-hour window was not completed in the retained execution and is reported as incomplete (Sections 4.4 and 6.1).

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Confirm the exact definition of a "vasopressor-active interval" used for $[s_i, e_i]$ — e.g., whether brief interruptions or titration-only periods count as continuously active, and whether this definition is consistent with the action definition in Section 3.6.]

3.11 Negative Controls and Sensitivity Analyses

A battery of negative-control analyses was pre-specified to help distinguish definitional/temporal leakage from treatment persistence, temporal autocorrelation, and ordinary clinical predictability, but had not completed execution at the time of this report (Experiment 5). Planned controls include:

- Shuffled-label controls: re-running the Experiment 2 recoverability probes after randomly permuting the action label within patient, to establish a chance-level performance floor.
- Temporal-offset analysis: re-running recoverability probes using action labels shifted by 0, 4, 8, 16, and 32 hours relative to the state window, to characterize how recoverability decays as temporal proximity to the action decreases — a pattern that would be consistent with treatment persistence/autocorrelation rather than, or in addition to, definitional/window-overlap leakage.
- Prevalence-matched placebo features: substituting the treatment-linked SOFA components with synthetic features matched on prevalence/marginal distribution but not mechanistically linked to treatment, to test whether Variant-A-level recoverability can be reproduced by features with similar statistical properties but no plausible leakage pathway.

These analyses are described here as planned methodology only; no results from Experiment 5 are reported, and none should be inferred from the completed experiments.

3.12 Statistical Analysis

Descriptive statistics (proportions, means) are reported for Experiments 1 and 3 as observed in the analytic dataset, without inferential testing beyond what is documented in the supplied results (i.e., no p-values or hypothesis tests are reported for these experiments in the source materials, and none are fabricated here). For Experiment 2, AUROC and mutual information are reported as point estimates for

each state variant and probe-model combination; formal confidence intervals were planned via clustered bootstrap (clustering by patient) but are not yet available for insertion.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert the finalized statistical analysis plan, including any correction for multiple comparisons across the fifteen variant-by-model combinations in Experiment 2, and confirm whether calibration and macro-F1 metrics — reportedly following the same broad ordering as AUROC — will be included with full supporting values or omitted if unavailable.]

4. Results

4.1 Cohort and Analytic Dataset Characteristics

Table 1. Cohort and analytic dataset characteristics.

Characteristic	Value
ICU stays (base cohort)	11,354
Combined vitals/labs/FiO2 rows	7,497,914
Vasopressor dose events	5,428,937
Timesteps scored, 4-hour window	280,837
Timesteps scored, 24-hour window	326,647
Timesteps in final Experiment 2 modeling table	170,299
Next-bin vasopressor action positive rate (Experiment 2 table)	44.6%

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert a cohort flow diagram (CONSORT-style) accounting for attrition from base cohort (11,354 stays) to scored timesteps (280,837 / 326,647) to the final Experiment 2 modeling table (170,299), including reasons for exclusion at each step. Insert baseline demographic and clinical characteristics of the cohort (age, sex, admission SOFA, ICU LOS, mortality) if collected; if not collected, state explicitly that this audit is a feature-level methodological study and baseline clinical characterization was out of scope.]

4.2 Experiment 1: Definitional Leakage Audit

Table 2. Dose determination of cardiovascular SOFA by aggregation window.

Aggregation window	% of all eligible timesteps dose-determined	% of vasopressor-treated timesteps dose-determined	Mean dose-attributable cardiovascular SOFA contribution (points)
4-hour (decision-aligned)	30.5%	100%	1.05
24-hour (conventional)	26.3%	100%	0.90

The population-level figures — 30.5% of all eligible timesteps under the 4-hour window and 26.3% under the 24-hour window — indicate that roughly a quarter to a third of all scored timesteps in this cohort carried a cardiovascular SOFA subscore that was mechanically shaped by vasopressor dosing rather than by MAP alone. The 100% figure among vasopressor-treated timesteps is consistent, by construction, with the scoring decomposition described in Section 3.7 (a treated patient's dose-based scoring band meets or exceeds their MAP-based band under the standard cardiovascular scoring rule) and is reported here as an internal-consistency check on the decomposition rather than as a novel substantive finding. The 4-hour window showed both a higher population-level dose-determined proportion and a higher mean dose-attributable point contribution than the 24-hour window.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert clustered-bootstrap 95% confidence intervals for the four proportions/means in Table 2.]

4.3 Experiment 2: Action Recoverability

Table 3. Action recoverability (next-bin vasopressor action) across state variants and probe models.

Variant	Description	Logistic regression AUROC	Random forest AUROC	Gradient boosting AUROC	Mutual information
A	Full conventional state	0.874	0.892	0.900	0.031
B	Without total SOFA	0.869	0.893	0.900	0.026
C	Without cardiovascular SOFA	0.845	0.855	0.885	0.023
D	Treatment-decomposed state	0.733	0.779	0.791	0.018
E	Physiology-only state	0.727	0.779	0.791	0.017

The ordering $A \approx B > C > D \approx E$ was consistent across all three probe-model families and was mirrored by the mutual information estimates. Two features of this ordering are noted descriptively here (interpretation is deferred to Section 5): first, removal of total SOFA alone ($A \rightarrow B$) produced minimal change in AUROC (≤ 0.005 across all three models) and only a modest reduction in mutual information ($0.031 \rightarrow 0.026$); second, the largest single-step reduction in recoverability occurred either at the removal of the cardiovascular SOFA subscore ($B \rightarrow C$: e.g., gradient boosting $0.900 \rightarrow 0.885$) or, more substantially, at the removal of the dose-attributable cardiovascular component specifically ($C \rightarrow D$: e.g., gradient boosting $0.885 \rightarrow 0.791$; logistic regression $0.845 \rightarrow 0.733$). Variants D and E produced nearly identical AUROC and mutual-information values across all three probe families (maximum difference 0.006 AUROC, logistic regression: 0.733 vs. 0.727; random forest and gradient boosting identical at 0.779 and 0.791 respectively).

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert clustered-bootstrap 95% confidence intervals for every cell in Table 3 (15 AUROC values + 5 mutual information values). Insert calibration metrics (e.g., Brier score, calibration slope) and macro-F1 for each variant/model combination if verified values become available; otherwise state explicitly that these are not reported due to unavailability rather than omitting mention of them. Insert results of statistical comparison between adjacent variants (e.g., DeLong test or bootstrap-based AUROC difference test) if performed.]

Figure 1 (caption). Action recoverability (AUROC) across nested state variants A–E, by probe model.

Line plot with state variant (A–E) on the x-axis and AUROC on the y-axis, with separate lines for logistic regression, random forest, and gradient boosting probes.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Generate and insert the actual figure from source data; the values needed to construct it are given in Table 3.]

4.4 Experiment 3: Temporal Window-Overlap Audit

Table 4. Temporal-overlap statistics, 24-hour conventional window.

Statistic	Value
Decision points evaluated	204,372
Mean contaminated-window fraction	56.1%
Median contaminated-window fraction	100%
25th percentile contaminated-window fraction	0%
% of decision points with >50% overlap with prior treatment period	55.7%

The distribution of the contaminated-window fraction was strongly bimodal-leaning at the extremes: a median of 100% overlap alongside a 25th percentile of 0% overlap indicates that a substantial subgroup of decision points had feature-aggregation windows either entirely uncontaminated by prior treatment or, conversely, entirely overlapping a prior treatment period, with 55.7% of all evaluated decision points exceeding 50% overlap. The corresponding calculation for the 4-hour decision-aligned window was not completed in the retained execution.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Complete and insert the 4-hour window contaminated-window-fraction results (mean, median, 25th percentile, % >50% overlap) to allow direct comparison against Table 4, since window length is a primary hypothesized driver of temporal/window-overlap leakage per Section 1. Insert a histogram or empirical CDF of the contaminated-window fraction (Figure 2) once source data permit.]

Figure 2 (caption). Distribution of contaminated-window fraction across 24-hour-window decision points.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Generate and insert histogram/CDF from source data.]

4.5 Status of Negative Controls (Experiment 5)

The pre-specified negative-control battery (shuffled-label controls, temporal-offset analysis at 0/4/8/16/32 hours, and prevalence-matched placebo features; Section 3.11) did not complete execution and no results are reported. This is stated here as a factual status report rather than as a limitation to be elaborated upon repeatedly elsewhere in the manuscript; its interpretive consequences are addressed once, in the Discussion and Limitations sections.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Insert Experiment 5 results in full (or, if the analysis is ultimately descoped rather than merely delayed, revise Section 3.11 and this subsection to state that explicitly, together with a justification).]

5. Discussion

5.1 Principal Findings

This audit produced three quantitative findings. First, a substantial minority of all SOFA-scored timesteps in this cohort — 30.5% under the 4-hour window and 26.3% under the 24-hour window — had a cardiovascular subscore that was mechanically shaped by vasopressor dosing rather than by measured MAP alone, establishing that definitional/construction-based treatment-information embedding is not a rare edge case but a routine feature of SOFA-based state construction in a vasopressor-exposed sepsis cohort. Second, the next-bin vasopressor action was substantially more recoverable from a full conventional state (AUROC 0.869–0.900 across probe families) than from a treatment-decomposed or physiology-only state (AUROC 0.727–0.791), with the largest single-step reduction in recoverability occurring around removal of the cardiovascular SOFA subscore and, more specifically, its dose-attributable component. Third, more than half of 24-hour-window decision points (55.7%) had feature-aggregation windows that overlapped a prior treatment period by more than 50%, indicating that temporal/window-overlap leakage is, at minimum, structurally plausible at scale in this cohort, independent of any scoring-definition mechanism.

Together, these findings indicate that a commonly used SOFA-based state representation for sepsis RL is measurably informative about the very treatment action it is meant to help decide, through at least one identifiable and quantifiable pathway (definitional construction) and one structurally documented but not yet fully isolated pathway (temporal overlap).

5.2 Interpretation of Definitional Leakage

The Experiment 1 results should be interpreted at two distinct levels, consistent with the distinction drawn in Section 3.8. The finding that 100% of vasopressor-treated timesteps were dose-determined is not, by itself, informative about the scale of the problem — it is a near-tautological consequence of the SOFA cardiovascular scoring rule, in which a treated patient's dose-based scoring band is defined to meet or exceed their MAP-based band. Its value in this manuscript is as an internal-consistency check confirming that the MAP-attributable/dose-attributable decomposition (Section 3.7) behaves as intended. The population-level figures — 26.3–30.5% of *all* eligible timesteps, not just treated ones — are the substantive finding: they quantify how much of the entire scored dataset, including untreated timesteps, carries a construction-based association with treatment status via this single subscore. The shorter, decision-aligned 4-hour window showed both a higher dose-determined proportion and a higher mean dose-attributable point contribution than the 24-hour window, a pattern discussed further in Section 5.6 in relation to window-length trade-offs.

5.3 Interpretation of Action Recoverability

The Experiment 2 results are consistent with, but do not on their own establish the magnitude of, treatment-information embedding via SOFA. Three specific patterns merit comment. First, the near-equivalence of Variants A and B (removing total SOFA while retaining subscores) suggests that total SOFA

contributes little *unique* recoverability once its component subscores remain in the state — an expected result given that total SOFA is an arithmetic function of the subscores already present in Variant B, and thus not independently informative beyond them. Second, the decline from B/A to C (removing the cardiovascular subscore specifically) indicates that the cardiovascular component carries recoverable, action-related signal beyond what the remaining subscores and raw physiological features provide. Third, the similarity between Variants D and E — where D retains the MAP-attributable cardiovascular component but removes the dose-attributable component, and E removes cardiovascular treatment-linked information more broadly — offers an internal consistency check suggesting that most of the treatment-linked recoverability captured by the cardiovascular subscore was concentrated in its dose-attributable component specifically, and that removing this component brought recoverability down close to what remains in a physiology-only representation.

None of these three observations, individually or jointly, licenses the conclusion that the full A-to-E AUROC gap (e.g., 0.900 to 0.791 for gradient boosting) is *caused by* leakage in a quantified sense. AUROC-based recoverability is a diagnostic signal of statistical association between state and action, not a causal estimate of how much of that association is attributable to a construction artifact as opposed to legitimate clinical predictability, treatment persistence, or temporal autocorrelation not yet isolated by the completed experiments. We therefore treat the observed gap as an **upper-bound estimate** on the magnitude of leakage-attributable recoverability, pending the negative-control analyses described in Section 3.11.

5.4 Interpretation of Temporal Overlap

The Experiment 3 findings establish that, under the 24-hour conventional window, the structural precondition for temporal/window-overlap leakage — overlap between a feature-aggregation window and a period of prior treatment — is common, affecting the majority of decision points by the >50%-overlap criterion. This is a necessary but not sufficient condition for temporal leakage to materially affect a given feature's value: overlap with a prior treatment period does not guarantee that the specific features aggregated over that window (e.g., MAP, lactate) were meaningfully altered by that treatment before the window closed. Because the corresponding 4-hour-window overlap analysis was not completed, we cannot yet report whether the decision-aligned window meaningfully reduces this structural exposure relative to the conventional window, despite the Experiment 1 result that the 4-hour window showed a *higher* dose-determined proportion for the cardiovascular subscore specifically. These two findings are not contradictory — Experiment 1 concerns the current-timestep scoring rule, whereas Experiment 3 concerns overlap with *prior* treatment — but their joint interpretation would benefit substantially from the pending 4-hour overlap comparison.

5.5 Distinguishing Leakage from Legitimate Clinical Predictability

A central interpretive challenge in this audit is that vasopressor administration is, appropriately, driven by illness severity: clinicians titrate vasopressors in response to hypotension and other markers of shock, and any valid clinical state representation should be expected to retain *some* ability to predict vasopressor use, because sicker patients legitimately receive more vasopressor support. A state representation with zero

action-recoverability would be difficult to reconcile with the premise that SOFA reflects clinically meaningful severity at all. The purpose of the nested-variant design is therefore not to drive recoverability to zero, but to characterize how much of the observed recoverability is specifically attributable to components that are treatment-linked *by construction* (cardiovascular SOFA's dose-attributable component) as opposed to components that reflect severity through physiological measurement alone (Variant E). The persistence of non-trivial recoverability even in Variant E (AUROC 0.727–0.791) is consistent with this expected legitimate predictability, and should not itself be interpreted as evidence of leakage. Equally, treatment persistence — the tendency for a vasopressor started in a recent bin to continue into the next bin, independent of any state-construction artifact — remains an uncontrolled explanation for at least part of the A-to-E gap until the temporal-offset negative control (Section 3.11) is completed, since persistence-driven and severity-driven predictability would both be expected to decay differently than a purely definitional or window-overlap artifact as the offset between state and action increases.

5.6 Implications for Offline Reinforcement Learning

If a state representation encodes information about the very action it is meant to inform, several downstream RL procedures may be affected in ways not directly evaluated by this audit. Policy-value estimation methods that rely on estimating the probability of observed actions given state (e.g., importance-sampling-based off-policy evaluation) implicitly depend on this conditional relationship, and a state representation with construction- or window-based associations with the action could distort such estimates in directions this study did not attempt to characterize. Similarly, a learned policy's recommendations at a given state could, in principle, partly reflect what a clinician *already did* rather than what should be done next, if treatment-linked information persists in the state used for policy training. We emphasize that this study did not train or evaluate a treatment policy (Experiment 4 was out of scope) and cannot speak to the direction or magnitude of any such downstream effect in any specific published sepsis-RL system; it establishes only that the structural precondition for such effects — recoverable treatment information in the state — is present and quantifiable in this cohort.

5.7 Implications for Representation Learning

The nested-variant methodology used in Experiment 2 offers a general template — decomposing a composite clinical score into its treatment-linked and non-treatment-linked components and measuring the marginal recoverability contributed by each — that could be applied to other composite severity scores used in clinical RL and prediction contexts beyond SOFA (e.g., other organ-dysfunction indices that incorporate treatment-dependent inputs). This template is diagnostic rather than corrective: it identifies where treatment-linked information is concentrated but does not, by itself, specify the optimal representation for downstream policy learning, which may involve trade-offs between removing treatment-linked information and preserving genuine severity signal.

5.8 Implications for SOFA-Based States and Rewards

Beyond its use as a state feature, SOFA (or its trajectory over time) is also commonly used as a component of reward functions in sepsis RL, on the premise that reducing SOFA reflects clinical improvement. Insofar as the cardiovascular subscore is partly treatment-determined, a SOFA-based reward could, in principle, credit a policy for changes in vasopressor dosing directly through the scoring rule rather than through an intervening physiological effect — a concern structurally analogous to, but distinct from, the state-representation leakage examined in this study, since reward-shaping was not directly audited here.

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: If the study team intends to make a direct claim about SOFA-based reward functions, insert a dedicated reward-focused analysis or explicitly scope this discussion as a hypothesis for future work rather than an audited finding.]

5.9 Implications for MIMIC-Based Research and Reproducibility

Because a substantial share of published sepsis-RL work has relied on MIMIC-III or MIMIC-IV and on SOFA-derived state representations, the mechanisms characterized here are structurally relevant across a broader literature than any single study. We do not, however, extend this audit's cohort-specific quantitative findings to other studies' cohorts, feature-engineering pipelines, or window conventions, which may differ from those used here in ways that materially change the magnitude — though likely not the qualitative existence — of the two leakage mechanisms described.

5.10 Recommendations for Future Studies

Based on the completed experiments, we recommend that future sepsis-RL studies using SOFA-derived states: (1) report the population-level dose-determined proportion for the cardiovascular subscore under their chosen aggregation window, analogous to Table 2; (2) report a nested-variant recoverability audit analogous to Table 3 as a standard diagnostic, particularly when the action space includes vasopressor dosing; (3) report window-overlap diagnostics analogous to Table 4 whenever feature-aggregation windows are used; and (4) complete negative-control analyses of the kind pre-specified in Section 3.11 before characterizing any observed recoverability gap in causal terms. We stop short of recommending a specific corrective state representation (e.g., mandating Variant-E-style physiology-only states) as a universal solution, since doing so would require the completed negative-control and, ideally, policy-sensitivity analyses that remain pending in this study.

6. Limitations

6.1 Limitations of the Empirical Audit

- Single-database, retrospective design. All findings are derived from a single institution's data (via MIMIC-IV) and a single retrospective cohort; generalizability to other institutions, EHR systems, or patient populations was not assessed.
- Dependence on MIMIC-IV data quality. Charting frequency, measurement error, and missingness patterns characteristic of retrospective EHR data may affect both the SOFA decomposition and the recoverability probes in ways not separately quantified in this audit.
- Creatinine-only renal SOFA implementation. The renal subscore was computed using creatinine alone, without urine output, which may cause this study's SOFA totals (used in Variants A and B) to differ somewhat from full standard SOFA; this is not expected to affect the cardiovascular-subscore-focused findings central to this manuscript's contribution but should be considered when comparing absolute AUROC values to other studies.
- Cohort-provenance confirmation pending. Several extraction and cohort-definition details (Sections 3.2–3.4) remain marked as placeholders pending confirmation and are essential to complete before submission for full reproducibility.
- Incomplete 4-hour temporal-overlap analysis. The Experiment 3 audit was completed only for the 24-hour window; the 4-hour-window comparison, which would directly test whether a shorter decision-aligned window reduces temporal-overlap exposure, remains outstanding.
- Incomplete negative-control battery. Experiment 5 (shuffled-label, temporal-offset, and placebo-feature controls) did not complete execution. Without these results, the Experiment 2 recoverability gap cannot be decomposed into leakage-attributable and non-leakage-attributable (treatment persistence, autocorrelation, legitimate predictability) components, and all Discussion-section causal language has been correspondingly hedged.
- Possible treatment persistence/autocorrelation. As discussed in Section 5.5, the observed recoverability gap may partly reflect the tendency of vasopressor administration to persist across consecutive bins, a phenomenon not yet isolated from construction- or window-based leakage.
- No external validation. All results pertain to this cohort and were not validated against an independent dataset.
- Action-definition limitations. As detailed in Section 3.6, several aspects of the next-bin vasopressor action's exact operationalization remain to be confirmed and documented; this limits precise interpretation of the 44.6% positive rate and of any dose-specific claims.
- No confidence intervals currently available. All Experiment 1–3 point estimates are reported without formal uncertainty quantification pending completion of clustered-bootstrap analyses; findings should accordingly be read as descriptive of this specific analytic sample rather than as precision-quantified population estimates.
- Cannot infer effects on specific published RL policies. This audit does not re-run or re-evaluate any previously published sepsis-RL study (e.g., Komorowski et al., 2018); it therefore cannot quantify how much, if at all, the leakage mechanisms characterized here would alter the specific reported results of any such study, and no such inference should be drawn from this manuscript.

- Reconstructed figures. Figures 1 and 2 (Sections 4.3–4.4) remain to be generated from underlying source data rather than from the summary statistics reported in the text; minor discrepancies could emerge during figure generation and should be reconciled before submission.

6.2 Limitations of the Literature Review

- Incomplete literature-audit table. The systematic per-study leakage-susceptibility assessment (Appendix A) could not be fully populated in this session because a separate conceptual/theoretical draft manuscript, referenced in the original task specification as a source for this review, was not supplied. Only Komorowski et al. (2018) has been discussed in any methodological depth; broader claims about “the sepsis-RL literature” in this manuscript should be read as provisional until this table is completed and verified against primary sources.
- Certainty categories are inherently constrained by published reporting. Even once populated, the literature-audit table's assessments will be limited by what each audited study's published Methods sections actually document; where methodological detail is not reported, this manuscript uses “cannot determine from published methods” rather than inferring a conclusion, which necessarily limits the strength of any literature-wide claim.
- No independent re-analysis of other studies' data. This review characterizes published descriptions of other studies' state/action constructions; it does not re-derive or verify those constructions from the underlying data each study used.

7. Conclusion

In an 11,354-stay MIMIC-IV sepsis cohort, a commonly used SOFA-derived clinical state representation was found to encode measurable information about a concurrent vasopressor treatment action through a mechanically documented definitional pathway — affecting 26–31% of all scored timesteps depending on aggregation window — and to be substantially more predictive of that action than a treatment-decomposed or physiology-only representation, with a structurally plausible, though not yet fully isolated, contribution from temporal overlap between feature-aggregation windows and prior treatment periods. These findings are consistent with meaningful treatment-information embedding in SOFA-based sepsis-RL state representations, but the precise magnitude attributable to leakage, as distinct from legitimate clinical predictability and treatment persistence, remains to be established through the negative-control analyses that this study pre-specified but did not complete. We recommend that state and reward construction in future sepsis-RL work be accompanied by explicit, reportable diagnostics of the kind introduced here — definitional decomposition audits, nested-variant recoverability audits, and window-overlap audits — as a matter of standard methodological practice, pending further work to translate these diagnostics into validated corrective representations.

References

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Additional references remain to be verified and added:

[CITATION NEEDED — verify source supporting: a primary causal-inference reference on post-treatment conditioning/collider bias (Section 2.6).]

[CITATION NEEDED — verify source supporting: primary methodological reference(s) on temporal aggregation bias in longitudinal EHR modeling (Section 2.7).]

[CITATION NEEDED — verify source supporting: MIMIC-III original database publication (Johnson et al., 2016), if this manuscript's final scope references MIMIC-III alongside MIMIC-IV; confirm exact volume/article number and DOI before use.]

[CITATION NEEDED — verify source supporting: any additional sepsis-RL studies to be added to Appendix A once the conceptual draft manuscript is supplied.]

Appendix A. Literature Audit of Sepsis Reinforcement-Learning Studies

This table is a placeholder pending receipt of the separate conceptual/theoretical draft manuscript referenced in the original task specification (see Stage 1 correspondence, Missing Information Item 1, and Limitations Section 6.2). Only one study — Komorowski et al. (2018) — is populated below, based on independently verified published methods. Additional rows should be added only after each cited study's methods have been individually verified against its primary publication.

Study	Dataset	State representation	Vasopress or action included?	SOFA / treatment-dependent severity feature included?	Temporal window reported?	Potential leakage mechanism	Certainty of assessment
Komorowski et al. (2018)	MIMIC-III	Vital signs, labs, and severity indices including SOFA-related components	Yes (discretized vasopress or dose, part of combined action space)	Yes (SOFA-related components reported as part of state)	Reported at a coarse level; exact aggregation window construction not fully specified in the published methods available to this audit	Potential susceptibility to both definitional and temporal/window-overlap mechanisms as characterized in this manuscript	Potential susceptibility (cannot fully determine mechanism-level detail from published methods alone)

[PLACEHOLDER — REQUIRED BEFORE SUBMISSION: Populate additional rows once the conceptual draft manuscript (or equivalent independently verified source material) is available. For each additional study, classify each cell using the three certainty categories specified in the study protocol: "Directly documented," "Potential susceptibility," or "Cannot determine from published methods." Do not characterize any study as having leakage as an established fact absent direct documentation in its published state/action construction.]